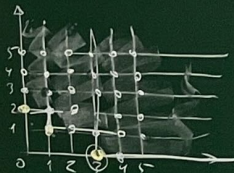


2.3.5 Lemma. Let $A \subseteq \mathbb{Z}_{\geq 0}^n$. Then $\langle x^\alpha : \alpha \in A \rangle = \lim_k \langle x^\beta : \beta \in A + \mathbb{Z}_{\geq 0}^n \rangle$.

2.3.5' Example. $I = \langle x^3, xy, y^2 \rangle$



$$\underbrace{(1+x+y)}_{\text{archaic}} x^3 + \underbrace{(1+y)}_{\text{archaic}} xy + \underbrace{(2-x)}_{\text{archaic}} y^2$$

$$3 \cdot xy^3 - x^2y + 7 \cdot x^5y + x^3y^4 = (7x^2y) x^3 + (3y^2 - x) xy + (x^3y^3) y^2$$

Proof: " $\supseteq$ ". If we have x^β with $\beta \in A + \mathbb{Z}_{\geq 0}^n$, then $\beta = \alpha + \gamma$ with $\alpha \in A$ and $\gamma \in \mathbb{Z}_{\geq 0}^n$ so that $x^\beta = x^{\alpha+\gamma} = x^\gamma \cdot x^\alpha$ belongs to the ideal. But since all x^β with $\beta \in A + \mathbb{Z}_{\geq 0}^n$ belong to our ideal, their k -linear combinations belong to the ideal.

" $\subseteq$ ". Assume $f \in \langle x^\alpha : \alpha \in A \rangle$. Then $f = \sum_{\alpha \in A} h_\alpha x^\alpha$, where $h_\alpha \in k[x_1, \dots, x_n]$ and $h_\alpha \neq 0$ for only finitely many choices of $\alpha \in A$. Each h_α is a k -linear combination of some monomials x^γ with $\gamma \in \mathbb{Z}_{\geq 0}^n$. When we expand brackets in $h_\alpha x^\alpha$, we obtain monomials $x^{\alpha+\gamma}$ with $\alpha+\gamma \in A + \mathbb{Z}_{\geq 0}^n$. So, by this, $f \in \lim_k \langle x^\beta : \beta \in A + \mathbb{Z}_{\geq 0}^n \rangle$. $\square$

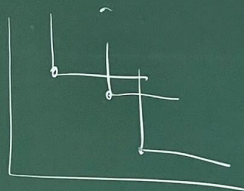
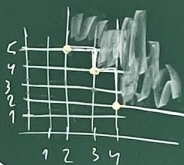
2.3.6. Corollary. For $A, B \subseteq \mathbb{Z}_{\geq 0}^n$ we have $\langle x^\alpha : \alpha \in A \rangle = \langle x^\beta : \beta \in B \rangle \iff$

(2)

$$A + \mathbb{Z}_{\geq 0}^n = B + \mathbb{Z}_{\geq 0}^n.$$

Proof: follows directly from 2.3.5.

2.3.7. Example. $\langle x^2y^5, x^3y^4, x^4y^2 \rangle$
 $\neq \langle x^2y^5, x^3y^4, x^4y^2, x^3y^3 \rangle$
 $= \langle x^2y^5, x^4y^2, x^3y^3 \rangle$

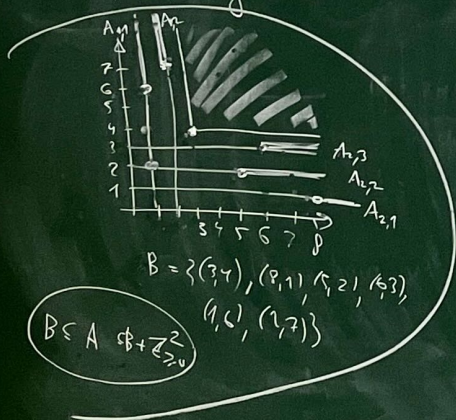


2.3.8 Theorem (Dickson's Lemma). For every $A \subseteq \mathbb{Z}_{\geq 0}^n$ there exists a finite $B \subseteq \mathbb{Z}_{\geq 0}^n$ with $\langle x^\alpha : \alpha \in A \rangle = \langle x^\beta : \beta \in B \rangle$.
 One can choose B to be a subset of A .

Proof: By corollary 2.3.6, it's enough to show that there exists a finite $B \subseteq A$ with $A + \mathbb{Z}_{\geq 0}^n = B + \mathbb{Z}_{\geq 0}^n$.
 Because $B \subseteq A$ this amounts to showing $A + \mathbb{Z}_{\geq 0}^n \subseteq B + \mathbb{Z}_{\geq 0}^n$. But for this inclusion to be true, it is enough to verify $A \subseteq B + \mathbb{Z}_{\geq 0}^n$. So let's prove this.

But this means that we want to show the existence of a finite $B \subseteq A$ such that for every $\alpha \in A$ there is some $\beta \in B$ with $\alpha \geq \beta$

(3)



The proof is by induction on n .

and we can assume that $A \neq \emptyset$ without loss of generality.

For $n=1$ the assertion is clear. We can take $B = \{\min A\}$.

Assume $n > 1$ and the assertion has been verified for subsets of $\mathbb{Z}_{\geq 0}^{n-1}$.

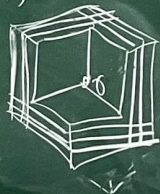
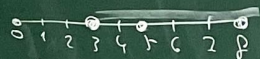
Pick an arbitrary $\gamma = (\gamma_1, \dots, \gamma_n) \in A$. So, for $\alpha \in A$, if

$\alpha \geq \gamma$ and γ is in B we are done. Otherwise

$\alpha_i < \gamma_i$ for some $i = 1, \dots, n$, which means $\alpha_i \in \{1, 2, \dots, \gamma_i - 1\}$ for some $i = 1, \dots, n$. It follows that

$$A \subseteq \{\alpha \in \mathbb{Z}_{\geq 0}^n : \alpha \geq \gamma\} \cup \bigcup_{i=1}^n \bigcup_{m=1}^{\gamma_i-1} A_{i,m} \quad \text{with } A_{i,m} = \{\alpha \in A : \alpha_i = m\}$$

Applying the induction hypothesis to $A_{i,m}$ we find a finite $B_{i,m} \subseteq A_{i,m}$ such that for every $\alpha \in A_{i,m}$ there is $\beta \in B_{i,m}$ with $\alpha \geq \beta$.



With this choice, we can fix $B = \{x^j\} \cup \bigcup_{i=1}^n \bigcup_{m=1}^{j_i-1} B_{i,m}$.

□ (4)

2.4. Hilbert basis theorem and Gröbner bases.

2.4.1 Def. For an ideal $I \subseteq k[x_1, \dots, x_n]$ with $I \neq \{0\}$, we define the initial ideal $LT(I)$ by

$$\langle LT(I) \rangle = \langle LT(f) : f \in I \setminus \{0\} \rangle = \langle LM(f) : f \in I \setminus \{0\} \rangle.$$

$\langle LT(I) \rangle$ depends on the choice of the underlying monomial order.

Remark. If $I = \langle f_1, \dots, f_s \rangle$, one doesn't necessarily have $\langle LT(I) \rangle = \langle LT(f_1), \dots, LT(f_s) \rangle$.

2.4.2. Example

$$f_1 = x^3 - 2xy$$

$$f_2 = x^2y - 2y^2 + x$$

Grlex order

$$x^2 \in \langle f_1, f_2 \rangle \text{ because } -y f_1 + x f_2 = -y(x^3 - 2xy) + x(x^2y - 2y^2 + x) = x^2$$

$$\langle LT(\langle f_1, f_2 \rangle) \rangle \ni x^2 \text{ but } x^2 \notin \langle LT(f_1), LT(f_2) \rangle = \langle x^3, x^2y \rangle$$



2.4.3. Proposition. Let $I \subseteq k[x_1, \dots, x_n]$ be an ideal with $I \neq \{0\}$. Then there exist finitely many polynomials $g_1, \dots, g_s \in I$ with $\langle LT(I) \rangle = \langle LT(g_1), \dots, LT(g_s) \rangle$.

(5)

Proof: follows from Dickson's lemma - every monomial ideal is finitely generated.

2.4.4. Theorem (Hilbert's basis theorem) Every ideal $I \subseteq k[x_1, \dots, x_n]$ is finitely generated, that is $I = \langle g_1, \dots, g_s \rangle$ for some $g_1, \dots, g_s \in I$.

Proof: If $I = \{0\}$, the assertion is trivial. Assume $I \neq \{0\}$. Let $g_1, \dots, g_s$ be finitely many polynomials in I with $\langle LT(I) \rangle = \langle LT(g_1), \dots, LT(g_s) \rangle$.

It turns out that $I = \langle g_1, \dots, g_s \rangle$. We need to show that each $f \in I$ belongs to $\langle g_1, \dots, g_s \rangle$.

We divide f by $g_1, \dots, g_s$ obtaining

$$f = a_1 g_1 + \dots + a_s g_s + r \quad \text{where } a_1, \dots, a_s, r \in k[x_1, \dots, x_n]$$

and no term of the remainder r divides any of $LT(g_1), \dots, LT(g_s)$.

$$\begin{aligned} 0 &\in I \\ I + I &\subseteq I \\ k[x_1, \dots, x_n] \cdot I &\subseteq I \end{aligned}$$